

A.13 RUNTIME TABLE

Table 17: **Runtimes.** We include the runtimes of both training per epoch (in seconds) and performing calibration. Reducing stochasticity can help reduce computation (L1 $\rightarrow$ L3). Cost can also be reduced by using a pretrained model.

Dataset	GOODCMNIST		GOODSST2		GOODMotifLPE	
	Training (S)	Inference (S)	Training (S)	Inference (S)	Training (S)	Inference (S)
Vanilla	18.5	25.8	10.8	18.5	3.8	4.5
Temp. Scaling	18.5	23.5	10.8	13.4	3.8	5.3
DEns (Ens Size=3)	18.456 x Ens Size	59.4	10.795 x Ens Size	29.0	3.8 x Ens Size	11.8
G- Δ UQ (L1, 10 anchors)	22.1	181.5	15.9	17.1	5.8	15.5
G- Δ UQ (L2, 10)	22.4	148.6	12.7	15.5	5.8	11.8
G- Δ UQ (HiddenRep, 10)	18.5	28.0	13.8	19.6	3.9	6.5
G- Δ UQ (Pretr. HiddenRep, 10)	8.6	27.8	6.8	16.0	2.5	6.4

A.14 MEAN AND VARIANCE OF NODE FEATURE GAUSSIANS

Table 18: **Mean and Variance of Node Feature Anchoring Gaussians.** We report the mean and variance of the Gaussian distributions fitted to the input node features. Because the input node features vary in size, we report aggregate statistics over the mean and variance corresponding to each dimension. For example, Min(Mu) indicates that we are reports the minimum mean over the d -dim set of means.

Dataset	Domain	Shift	Min (Mu)	Max (Mu)	Mean (Mu)	Std (Mu)	Min (Std)	Max (Std)	Mean (Std)	Std (Std)
GOODSST2	length	concept	-4.563	0.69	-0.011	0.278	0.163	0.803	0.242	0.049
GOODSST2	length	covariate	-4.902	0.684	-0.01	0.3	0.175	0.838	0.255	0.05
GOODCMNIST	color	concept	0.117	0.133	0.127	0.008	0.092	0.097	0.095	0.003
GOODCMNIST	color	covariate	0.087	0.131	0.102	0.025	0.108	0.109	0.108	0
GOODMotifLPE	size	covariate	0.003	0.021	0.011	0.008	0.835	1.728	1.248	0.377
GOODMotifLPE	size	concept	-0.006	0	-0.002	0.003	0.542	1.114	0.783	0.242
GOODMotifLPE	basis	concept	-0.011	0.015	0.001	0.011	0.721	1.464	1.09	0.304
GOODMotifLPE	basis	covariate	-0.007	-0.002	-0.004	0.002	0.808	1.913	1.251	0.469
GOODWebKB	university	concept	0	0.95	0.049	0.099	0.001	0.5	0.168	0.095
GOODWebKB	university	covariate	0	0.934	0.05	0.104	0.001	0.5	0.164	0.098
GOODCora	degree	concept	0	0.507	0.007	0.017	0.001	0.5	0.061	0.051
GOODCora	degree	covariate	0	0.518	0.007	0.017	0.001	0.5	0.061	0.052
GOODCBAS	color	covariate	0.394	0.591	0.471	0.093	0.142	0.492	0.403	0.174
GOODCBAS	color	concept	0.23	0.569	0.4	0.144	0.168	0.495	0.39	0.152

GOODCMNIST Color, Concept: Anchoring Distribution vs. Input Features

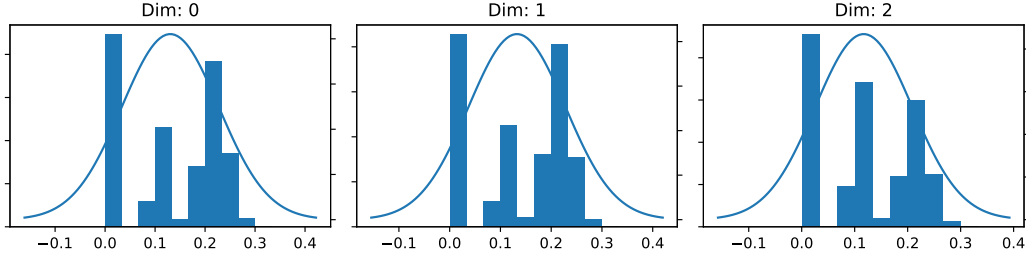


Figure 9: **GOODCMNIST, Concept, Anchoring Distribution.** We plot the mean and variance of the fitted anchoring distribution vs. the true feature distribution for each input dimension. We observe there is a mismatch between the empirical distribution and the fitted Gaussian. However, we did not find this mismatch to harm the effectiveness of G- Δ UQ.

GOODCMNIST Color, Covariate: Anchoring Distribution vs. Input Features

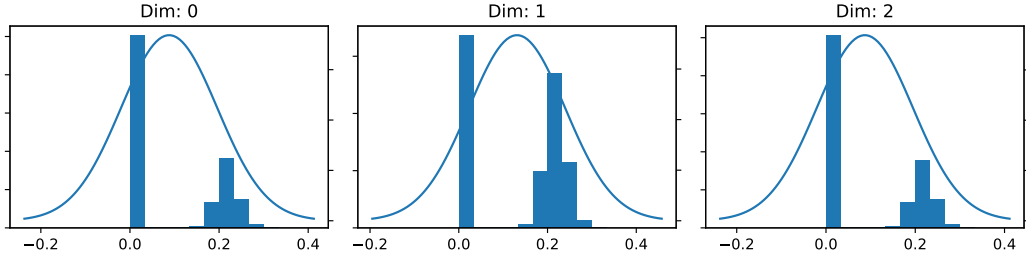


Figure 10: **GOODCMNIST, Covariate, Anchoring Distribution.** We plot the mean and variance of the fitted anchoring distribution vs. the true feature distribution for each input dimension. We observe there is a mismatch between the empirical distribution and the fitted Gaussian. However, we did not find this mismatch to harm the effectiveness of G- Δ UQ.

Table 19: **Number of Parmeters per Model.** We provide the number of parameters in the vanilla and modified parameter as follows. Note, that the change in parameters is architecture and input dimension dependent. For example, GOODCMNIST, and GOODSST2 use GIN MPNN layers. Therefore, when changing the layer dimension, we are changing the dimension of its internal MLP. It is not an error that intermediate layer G- Δ UQ have the same number of parameters, this is due to the architecture: these layers are the same size in the vanilla model. Likewise, GOODCora’s input features have dimension is 8701, so doubling the input layer’s dimension appears to add a significant number of parameters. We do not believe this

Dataset	GOODCMNIST	GOODMotif	GOODSST2	GOODCORA	GOODWebKB	GOODCBAS
Baseline	2001310	911403	1732201	2816770	695105	185104
G- Δ UQ(NFA)	2003110	913803	2193001	5429770	1206005	186304
G- Δ UQ(L1)	2360110	1633203	2091001			
G- Δ UQ(L2)	2360110	1633203	2091001			
G- Δ UQ(L3)	2360110					
G- Δ UQ(L4)	2360110					
G- Δ UQ(Readout)	2004310	912303	1732501			